

SL_No	Research Paper Title	Link	Survey	Student Name
1	Machine Learning for Seed Quality Classification: An Advanced Approach Using Merger Data from FT-NIR Spectroscopy and X-ray Imaging	https://www.mdpi.com/1424-8220/20/15/4319	Recent research on seed quality reveals that ML and DL approaches continuously achieve high accuracy rates which ranges from 85-99% across a variety of imaging and spectroscopy techniques. Review of 10 peer-reviewed studies highlights the strength of methodologies like CNN-based deep learning models which reached 99% accuracy for multi-seed classification and 95% for maize seed defects detection, while X-ray imaging paired with ML achieved 90% accuracy for germination prediction in forage grass and 94.36% for <i>Jatropha curcas</i> seed viability. X-ray CT imaging along with pattern recognition obtained 88.7% accuracy for pepper seed classification and FT-NIR spectroscopy alone reached 82% improving to 85% when combined with X-ray data. All together these non-destructive, automated approaches outperform traditional visual inspection methods across multiple species of crop and its quality parameters. One particularly novel contribution is a combination of FT-NIR spectroscopy and X-ray imaging for seed classification based on quality, representing one of the first attempts to integrate these two modalities. For germination prediction, FT-NIR achieved 82% X-ray 90% and the combined approach of 85% while seed vigor prediction proved more challenging with accuracies of 61%, 68% and 62% respectively, which reflected the difficulties of capturing complex influencing factors such as the environmental conditions. Though there were several promising results, there still remained many research gaps. Exploration on how combining datasets can enhance classification performance and the methodology requires validation across additional species beyond <i>Urochloa brizantha</i> to establish broader generalizability.	Delna
2	A Convolution Neural Network-Based Seed Classification System	https://www.mdpi.com/2073-8994/12/12/2018	A CNN-based seed classification system proposed by the authors addresses key gaps in the existing literature, including the limited application of deep learning techniques for seed classification, the predominance of traditional machine learning approaches, and the scarcity of studies handling subject heterogeneity across multiple seed varieties. To overcome these limitations, the system employs a Convolutional Neural Network with transfer learning, symmetry application, image augmentation, decayed learning rate, and model checkpointing to efficiently classify 14 commonly known seed types. The model was evaluated on a test set of 234 images, achieving a classification accuracy of up to 99.9%, and outperformed conventional machine learning models across key performance metrics including precision, recall, and F1-score for each seed class.	Anantha
3	Seeds Classification and Quality Testing Using Deep Learning and YOLO v5	https://dl.acm.org/doi/10.1145/3484824.3484913	Gulzar et al. (2020) developed a DL based system called the Mixed Cropping Seed Classifier and Quality Tester(MCSCQT), which was designed to address two largely unexplored challenges in seed science. Those were classifying seeds from mixed cropping systems and distinguishing seed of varying quality standards. The system was trained on a curated dataset of labelled images encompassing both healthy and diseased seeds of pearl millet and disease status within maize seeds. MCSCQT achieved an impressive precision and recall of 99%, demonstrating that deep learning can deliver near-perfect classification performance even in complex, multi-class scenarios. This work is significant because prior models had shown limited capability in handling mixed cropping seed classification, and the field lacked robust frameworks for quality-based seed discrimination. By directly targeting these gaps, MCSCQT represents a meaningful step forward in automating seed quality assessment, though further refinement and testing across a broader range of crop species and growing conditions would be needed to fully generalize its application.	Delna
4	Deep transfer learning based photonics sensor for assessment of seed-quality	https://www.sciencedirect.com/science/article/abs/pii/S0168169922002083?	Addressing the inefficiencies and unreliability of conventional seed quality assessment methods, this study introduced a photonics sensor powered by deep transfer learning, combining laser backscattering with architectures including VGG16, VGG19, ResNet50, and InceptionV3. InceptionV3 emerged as the strongest performer at 98.31% accuracy, with results validated against agronomic quality parameters such as percentage imbibition, radicle length, and germination percentage, all confirmed statistically significant at $p < 0.05$.	Anantha
5	Emerging thermal imaging techniques for seed quality evaluation: Principles and	https://www.sciencedirect.com/science/article/abs/pii/S0963996920300508?	Thermal imaging has emerged as a promising non-contact approach to seed quality evaluation, operating on the principle of detecting infrared radiation emitted by seeds to draw meaningful conclusions about their condition. A comprehensive review of the technique reveals its versatility across a range of quality assessment tasks, including the estimation of seed viability, detection of fungal growth and insect infections, identification of physical damage and impurities, seed classification, and variety identification, all done without physically interfering with the seeds themselves. In terms of performance, thermal imaging has proven effective at distinguishing viable from non-viable seeds, detecting biological infections with high accuracy, and achieving classification accuracy rates between 76% and 95%, making it one of the more reliable non-destructive and rapid methods available for seed quality evaluation. However, the technology is not without its limitations. Current thermal imaging systems can be costly and are sensitive to environmental conditions, which constrains their practical deployment at scale. There also remain meaningful gaps in detector design and image processing algorithms that need to be addressed before the technique can be widely adopted in industrial settings. Perhaps most critically, the development of low-cost yet accurate infrared detectors remains an open challenge, one whose resolution could significantly broaden access to this technology and accelerate its integration into routine seed quality assessment workflows.	Delna
6	Assessment of seed quality using non-destructive measurement techniques	https://www.cambridge.org/core/journals/seed-science-research/article/assess	A comprehensive review of non-destructive seed quality assessment techniques revealed that methods such as machine vision, spectroscopy, hyperspectral imaging, thermal imaging, electronic nose, and soft X-ray imaging hold considerable promise for accurate and reliable seed classification. Each technique proved capable of addressing specific quality parameters, ranging from disease detection and variety identification to chemical composition analysis and microbial pathogen detection, without compromising the integrity of the seed sample. However, the review also brought to light several practical challenges that continue to hinder large-scale implementation, including prohibitive instrumentation costs, heavy dependence on environmental conditions, extensive data requirements, and time-intensive processing pipelines. These findings collectively underscored the urgent need for faster, more robust instrumentation and refined analytical frameworks to translate the demonstrated potential of non-destructive techniques into consistent, field-ready solutions for seed quality monitoring.	Anantha
7	Integration of machine learning in seed quality analysis	https://www.biochemjournal.com/special-issue/2025.v9.i11S.1.6313/integrat	This study investigated the growing role of machine learning in modernising seed quality analysis, responding to the well-documented shortcomings of conventional testing methods, which remain destructive, labour-intensive, and highly prone to human error. A broad spectrum of ML models including Support Vector Machines, Random Forests, Naïve Bayes, K-Nearest Neighbours, Artificial Neural Networks, and Convolutional Neural Networks were integrated with advanced imaging technologies such as multispectral, hyperspectral, terahertz, X-ray, and RGB imaging, yielding strong performance in seed classification, defect detection, vigour estimation, and variety identification. The adoption of AI-powered platforms like SeedGerm and AIseed further extended these capabilities by enabling high-throughput, real-time analysis for germination prediction, moisture estimation, and physical purity discrimination. Collectively, the results positioned machine learning as a transformative and non-destructive alternative that substantially improves both the efficiency and accuracy of seed quality assessment.	Anantha
8	Precision seed certification through machine learning	https://www.maxapress.com/article/doi/10.48130/tia-0024-0013	A comprehensive review of machine learning applications in seed quality classification highlights how the combination of ML algorithms and optical sensors has transformed the way seeds are assessed, offering reliable, accurate, and considerably faster evaluations compared to conventional methods. The reviewed studies draw on a broad spectrum of techniques, ranging from classical classifiers such as K-means, Support Vector Machines (SVM), Discriminant Analysis (DA), Naïve Bayes (NB), and Random Forest (RF), to more sophisticated approaches including Artificial Neural Networks (ANNs) and Convolutional Neural Networks (CNNs). Collectively, these methods have demonstrated strong performance in seed quality assessments, and the continued progress in deep learning has even enabled the development of smartphone applications capable of discriminating between seed varieties and a development that signals a shift toward more accessible and field-deployable tools. Underlying this body of work is a clear acknowledgment that conventional seed quality assessment methods fall short in terms of efficiency and cost-effectiveness, and that there is a growing need for technologically advanced alternatives. Machine learning, in this context, is not merely an academic exercise but a practical response to real limitations in the industry, with the ongoing challenge being to make these approaches more scalable, affordable, and widely adoptable across diverse agricultural settings.	Delna
9	Hyperspectral imaging for seed quality and safety inspection	https://link.springer.com/article/10.1186/s13007-019-0476-y	Targeting the classification of maize seed defects, this study highlighted the inadequacy of traditional machine learning methods that suffer from repetitive feature extraction processes and limited classification accuracy. CNNs paired with transfer learning were adopted as the primary framework, with GoogleNet benchmarked against the conventional SURF+SVM method. GoogleNet achieved a classification accuracy of 95% compared to 79.2% from the traditional approach, establishing a substantial performance gap in favour of deep learning. Network depth was also found to positively influence accuracy, further validating the architectural choices made. The use of visualisation tools such as feature maps and heat maps added an interpretable layer to the model outputs, making the approach not only accurate but also analytically transparent and well-suited for integration into automated seed manufacturing workflows.	Anantha
10	Recent Applications of Multispectral Imaging in Seed Phenotyping and Quality Monitoring—An Overview	https://www.mdpi.com/1424-8220/19/5/1090	Multispectral imaging has gained considerable attention as a powerful tool for seed quality assessment and phenotyping, with a growing body of published research demonstrating its effectiveness across a range of fully-operated systems. The technique draws on multiple imaging modalities which also included spectral, thermal, fluorescence, X-ray, and magnetic resonance imaging which is typically paired with LED illumination, chemometric data analysis, high-speed digital image detection, and multivariate analyses to extract meaningful quality information from seeds in a non-destructive manner. The results reported across studies are notably strong: multispectral imaging with multivariate analyses has proven capable of evaluating seed purity and varietal discrimination, distinguishing viable from dead seeds with accuracies of 92% and 96%, and classifying different seed varieties with accuracy rates ranging from 83% to 100%. The technique has also shown promise in detecting aflatoxin contamination, achieving 92.3% accuracy for red peppers and 79.17% for hazelnut kernels, as well as classifying sunflower seeds by rancidity levels using PCA-CA, with relevant spectral wavelengths tied to biochemical markers such as carotenoids, anthocyanins, fat, and water content. Despite these encouraging results, the field is still navigating several significant challenges. There remains a lack of unified definition for seed quality, and most research focuses on batch-level rather than individual seed assessment, making it difficult to capture variation among single seeds. Traditional methods continue to be hampered by their destructive and labor-intensive nature, while the implementation of spectral imaging for real-time applications is constrained by the absence of a universal hyperspectral image analysis algorithm, the need for careful calibration, sensitivity to environmental factors, limited commercial availability of viable systems, and the ongoing challenge of selecting optimal wavelengths for practical deployment.	Delna
11	Seed Image Analysis and Its Application in Seed Science Research	https://juniperpublishers.com/aibm/AIBM_MS_ID.555709.php	Seed quality classification covers a range of critical criteria like purity, germination, moisture, and disease freedom and image analysis has established itself as a reliable method for assessing cultivar identity and distinctness by capturing visual and morphological traits that closely correlate with genetic properties. The diversity of techniques applied across the literature reflects how seriously researchers have taken this challenge, drawing on tools ranging from Smart Grain phenotyping software and frequency domain image analysis to texture and shape neural networks, X-ray imaging, K-mean algorithms, and QTL analysis. The results speak to the breadth of the field: Venora et al. identified Italian bean landraces, Yan et al. determined maize seed purity from color alone with recognition rates exceeding 93.75%, and Han et al. pinpointed stress cracks in corn kernels through frequency domain analysis. Vigor assessment has been equally well explored, with Sako et al. using automated imaging for lettuce, Li et al. building a visual detection system for vegetable seeds, and Francisco and Silvio linking mechanical damage in sweet corn directly to germination and vigor outcomes. Disease detection has also advanced considerably, with Kinnari et al. identifying seed-borne diseases in soybean, Alenear et al. evaluating pumpkin seeds infected with <i>ScMV</i> , and Vresak et al. distinguishing infected from healthy seed parts in winter wheat and triticale. However, the field still has ground to cover particularly in computing technology, pattern recognition, and data processing while cost-effective hardware and software solutions and scalable online sorting and grading systems remain priorities that the research community has yet to fully address.	Delna

12	An Efficient Method for Seeds Classification and Quality Testing Using Fast R-CNN	https://ieeexplore.ieee.org/document/10220730	Traditional seed sorting has long relied on manual visual inspection by experts which is a process that is not only time-consuming and labor-intensive but also prone to human error, making it ill-suited for the demands of modern agriculture where seed quality directly influences crop output. Addressing this gap, researchers proposed an automated seed classification and quality testing approach built around a Fast R-CNN model, designed to bring both speed and accuracy to a problem that conventional methods have struggled to handle efficiently. The system operates through a carefully structured five-stage computational pipeline that begins with Wiener filter preprocessing to reduce noise, followed by histogram equalization for image enhancement, binarization, and adaptive watershed segmentation to isolate individual seeds, before finally feeding the processed images into the Rapid R-CNN model for classification. When benchmarked against other deep learning architectures, including CNN and RNN models, the proposed approach demonstrated superior performance, establishing Fast R-CNN as a more effective choice for this application. While the abstract does not detail specific numerical metrics, the directional finding is clear that this framework represents a meaningful step toward fully automated seed sorting systems that can reduce human dependency, improve classification consistency, and ultimately contribute to better quality control in agricultural production pipelines.	Delna
13	Multispectral and X-ray images for characterization of Jatropha curcas L. seed quality	https://link.springer.com/article/10.1186/s13007-021-00709-6	Conventional seed quality evaluation methods such as visual inspection and germination tests have long been criticised for being destructive, labour-intensive, time-consuming, and heavily dependent on the subjectivity of trained analysts. To address these limitations, this study proposed a non-destructive classification framework that combined multispectral and X-ray imaging technologies with Linear Discriminant Analysis to objectively assess seed quality in Jatropha curcas L. The approach demonstrated consistently high predictive accuracy, with reflectance at 940 nm and X-ray data individually achieving accuracies exceeding 0.96, and the combined use of both modalities pushing predictive performance to a range of 0.96 to 0.98 across multiple quality traits. The strong correlation observed between imaging data and seed physiological performance confirmed that multispectral and X-ray imaging together constitute a reliable, rapid, and sustainable alternative to traditional seed quality assessment methods.	Anantha
14	Analysis and prediction of seed quality using machine learning	https://jece.iaescore.com/index.php/IJCE/article/view/28528	Motivated by the absence of reliable, technology-driven approaches in seed quality prediction, this study developed a CNN-based model to categorise seeds into premium, standard, or regular quality grades, aiming to support higher crop yields and consistent harvests. The model was trained on a dedicated seed dataset, achieving a training accuracy of 97%, though the testing accuracy of 64% revealed a notable generalisation gap, pointing to challenges such as limited dataset availability and insufficient validation frameworks. Supplementary experiments incorporating FT-NIR and X-ray data offered additional perspective, with X-ray data outperforming FT-NIR across classification and prediction tasks, achieving 90% and 68% accuracy respectively, while the combined use of both modalities yielded 85% accuracy for germination prediction and 62% for seed vigour estimation. Overall, the study underscored both the promise and the current limitations of CNN-based seed quality prediction, particularly the pressing need for larger, more diverse datasets and stronger validation methodologies to improve real-world reliability.	Anantha
15	Seed Quality Prediction using Computer Vision and Convolutional Neural Networks	https://ieeexplore.ieee.org/document/9985705	Addressing the persistent inefficiencies of manual seed quality inspection, which is not only labour-intensive but also prone to human error, this study developed a computer vision-based model using OpenCV and Convolutional Neural Networks to automate the prediction of seed quality. The model was designed to assess seed lots and generate purity percentage predictions, effectively reducing the dependence on human intervention in the quality screening process. By automating the detection and filtering of damaged seeds, the approach demonstrated a practical pathway toward faster, more consistent, and error-free seed quality assessment. The study reinforced the value of integrating computer vision with deep learning as a scalable solution for replacing manual inspection workflows in agricultural seed processing.	Anantha
16	Research on Classification Method of Maize Seed Defect Based on Machine Vision	https://onlinelibrary.wiley.com/doi/10.1155/2019/2716975	The classification of maize seed defects has traditionally depended on handcrafted feature extraction methods based on color, shape, and texture approaches that, while intuitive, are labor-intensive, lack scalability, and fall short in accuracy when applied to the nuanced visual differences that distinguish defective from healthy seeds. Huang et al. tackled this limitation head-on by evaluating deep learning as an alternative, specifically leveraging convolutional neural networks (CNNs) and transfer learning through the GoogLeNet architecture to automate the defect classification process. The results were striking: GoogLeNet achieved an accuracy of 95%, compared to just 79.2% for the traditional SURF+SVM approach, and the study further showed that classification performance improved as network depth increased, a finding that reinforces the value of deeper architectures for complex visual tasks. Beyond raw accuracy, the researchers employed visualization technologies such as feature maps and heat maps to shed light on how the model arrives at its decisions, addressing a common criticism of deep learning around interpretability. This combination of high accuracy, improved transparency, and compatibility with automated seed manufacturing pipelines positions CNNs and transfer learning as genuinely transformative tools in seed quality assessment and highlights just how significant the gap remains between what traditional machine learning methods can offer and what modern deep learning architectures are capable of delivering at scale.	Delna
17	Classification of maize seeds of different years based on hyperspectral imaging and model updating	https://www.sciencedirect.com/science/article/abs/pii/S0168169916300	One of the less-discussed but genuinely important challenges in seed quality classification is the problem of temporal variability and the fact that seeds harvested in different years can exhibit measurable differences in their spectral properties, which can quietly erode the accuracy of classification models built on historical data. Addressing this, researchers applied hyperspectral imaging across the 400 nm to 1000 nm reflectance range to evaluate 2000 maize seeds spanning four varieties and multiple harvest years, developing classification models based on mean spectral features using a least squares support vector machine (LSSVM). Recognizing that a static model would struggle to remain accurate as seed characteristics shift over time, the team introduced an incremental support vector data description approach to enable online model updating which essentially allows the system to adapt dynamically rather than becoming outdated. The impact was substantial: the updated LSSVM model achieved a classification accuracy of 94.4%, outperforming non-updated models by a notable margin of 10.3%. This finding makes a compelling case not just for hyperspectral imaging as a tool for maize seed classification, but more broadly for the importance of building adaptability into machine learning pipelines used in agricultural quality assessment, a consideration that has received limited attention in the literature but clearly has meaningful consequences for real-world performance.	Delna
18	Research on Maize Seed Classification and Recognition Based on Machine Vision and Deep Learning	https://www.mdpi.com/2077-0472/12/2/232	Classifying maize seeds accurately is far from straightforward; it is the growing number of varieties, overlapping color characteristics, and the inevitable mixing of seeds during production all combine to make reliable identification a genuine challenge, one that traditional methods have consistently struggled to meet in terms of both accuracy and real-time applicability. To address this, researchers developed a rapid classification system using machine vision and deep learning, building a dataset of 8080 seeds drawn from five distinct maize varieties and benchmarking a range of CNN architectures against one another. The results were impressive across the board: AlexNet reached 97.91%, GoogLeNet 97.84%, MobileNet 98.58%, and EfficientNet 98.28%, but it was the custom P-ResNet model which is trained with the Adam optimizer and transfer learning, that stood out, achieving an overall accuracy of 99.70% with variety-specific accuracies ranging from 99.61% for LiaoGe to 99.80% for KouXian, all while maintaining a model loss hovering around just 0.01. These figures represent a meaningful leap over what conventional approaches can deliver and demonstrate that deep learning-based machine vision is well-suited for high-throughput, real-time seed classification. That said, the study did caution about its boundaries and the dataset, while substantial, is limited in variety, and the model has not yet been tested against the additional variability introduced by differences in planting year, growing region, and climate conditions. Optimizing model stability across these dimensions remains an open and important area for future work.	Delna
19	Efficient Classification of Maize Seed Varieties Using Optimized Deep Learning Techniques	https://ieeexplore.ieee.org/document/11208860	Traditional manual approaches to maize seed classification have long been criticised for their slow throughput and susceptibility to error, creating a clear demand for more efficient and automated alternatives. This study responded to that need by developing SwiftNet-VGG, a deep learning system built on an optimised VGG architecture with transfer learning, complemented by preprocessing steps including image resizing, data augmentation, and input normalisation. The model achieved a validation accuracy of 92.56% and a macro-average F1-score of 0.91, reflecting strong and consistent classification performance across maize seed varieties. The results positioned SwiftNet-VGG as a faster and more dependable solution compared to conventional manual methods, contributing a practical and scalable framework for automated seed variety classification in quality assessment workflows.	Anantha
20	Machine learning approaches for grain seed quality assessment: a comparative study	https://link.springer.com/article/10.1007/s42452-025-07191-y	Conducted against the backdrop of a serious agricultural challenge in Malawi, where time-consuming testing methods and the widespread circulation of fake and low-quality seeds continue to undermine food security and farmer productivity, this study evaluated five machine learning algorithms including K-Nearest Neighbour, Logistic Regression, Naïve Bayes, Random Forest, and CNN on a dataset of 2460 maize seed samples. K-NN and Logistic Regression emerged as the standout performers, achieving perfect scores of 100% across accuracy, precision, recall, and F1-score, while CNN followed closely with 92% accuracy. A particularly striking finding was that 46.2% of the sampled seeds were classified as poor quality, largely attributed to improper handling practices rather than genetic factors. The study recommended K-NN and Logistic Regression as the most suitable algorithms for grain seed classification on well-structured agricultural datasets, while also acknowledging the need for broader dataset diversity to strengthen the generalisability of these findings.	Anantha
21	PREDICTION OF RANKING OF LOTS OF CORN SEEDS BY ARTIFICIAL NEURAL NETWORKS	https://www.scielo.br/j/eagri/a/9MGZWgXmdszwnGHWYXHf8bcv/?lang=en	The seed industry has long grappled with the challenge of processing large volumes of physiological quality data efficiently, with traditional decision-making processes proving too slow and inconsistent to meet the demands of modern seed lot evaluation. This study addressed that bottleneck by applying a range of machine learning algorithms including J48, Random Forest, Classification via Regression, lazy J48, Multilayer Perceptron, and Naïve Bayes through the Weka software platform, using data collected from eight maize seed crops. The models demonstrated high precision in classifying seed lots, with vigour attributes emerging as particularly influential predictors in determining lot rankings. The findings established that machine learning-driven analysis of vigour test data can meaningfully enhance the speed and accuracy of seed lot classification, offering the seed sector a more reliable and scalable framework for physiological quality assessment and decision-making.	Anantha
22	Machine learning approach for the classification of corn seed using hybrid feature selection	https://www.tandfonline.com/doi/full/10.1080/10942912.2020.1778723	Recognising the limitations of manual corn seed classification and the growing need for automated, accurate alternatives, this study proposed a hybrid feature-based machine learning framework that combined histogram, texture, and spectral features with correlation-based feature selection and threshold-based region growing segmentation. Four classifiers were evaluated including Random Forest, BayesNet, LogitBoost, and Multilayer Perceptron, with MLP delivering the strongest overall performance at 98.93% accuracy on regions of interest sized 150×150 pixels. Variety-level results were equally impressive, with Neelam Makkaï achieving the highest accuracy at 99.9%, followed by Desi Makkaï at 98.8%, ICI 339 at 99.4%, Kashmiri Makkaï at 98.5%, Pioneer P-1429 at 98.6%, and Sygenta ST-6142 at 97%. The study also acknowledged the influence of camera resolution and seed variety modifications on classification outcomes, pointing to the importance of comprehensive and diverse datasets for broader model validation and generalisation.	Anantha
23	Identification of Maize Seed Defects using the Yolo-V5 Algorithmic Framework	https://journals.ku.ac.ke/index.php/AJPAS/article/view/740	With manual seed quality testing remaining a persistent source of inefficiency and error, and the increasing pressure on agricultural systems due to land degradation demanding more effective sorting technologies, this study explored the application of object recognition paradigms and deep learning models for automated maize seed defect identification. YOLOv5 and YOLOv8 were evaluated as the primary architectures, with YOLOv5 achieving an F1-score of 97.7% and YOLOv8 delivering a precision of 98.4% and a mean average precision of 98.2%, culminating in an overall classification accuracy of up to 98%. The results demonstrated that YOLO-based frameworks are highly capable of detecting and classifying seed defects with speed and precision, presenting a compelling case for the adoption of advanced computer vision technology.	Anantha

24	Application of Hyperspectral Imaging and Chemometric Calibrations for Variety	https://www.mdpi.com/1424-8220/12/12/17234	Traditional methods for maize seed variety identification have consistently fallen short in terms of speed, cost-efficiency, and reliability, creating a clear need for more objective and technically advanced alternatives. This study addressed that gap by applying VIS-NIR hyperspectral imaging combined with texture analysis and chemometric calibrations, utilising Principal Component Analysis, Kernel PCA, and Least Squares Support Vector Machine to discriminate between maize seed varieties. The PCA-GLCM-LS-SVM model emerged as the most effective configuration, achieving a recognition accuracy of 98.89% and demonstrating that the integration of spectral and textural information yields highly discriminative features for variety classification. The findings established hyperspectral imaging as a rapid, non-destructive, and accurate tool for maize seed variety discrimination, offering a technically sound alternative to the time-consuming and error-prone conventional approaches that have long dominated this domain.	Anantha
25	Maize Seed Variety Classification Using the Integration of Spectral and Image F	https://www.mdpi.com/2076-3417/6/6/183	Prior approaches to hyperspectral imaging-based seed classification were hindered by slow processing speeds, high-dimensional feature spaces prone to the curse of dimensionality, and comparatively limited accuracy, with earlier models plateauing at 83.68% and 76.18% for combined and spectral-only features respectively. This study tackled these gaps by introducing a novel pipeline that integrated spectral and image features extracted through hyperspectral imaging, applied the Successive Projections Algorithm for optimal wavelength selection to improve acquisition and processing efficiency, and employed Principal Component Analysis and Multidimensional Scaling for feature transformation and dimensionality reduction. The resulting LS-SVM model achieved a test accuracy exceeding 90%, representing a meaningful improvement over previous approaches and validating the benefit of combining feature integration with transformation techniques. The work demonstrated that addressing processing speed, feature dimensionality, and classification accuracy in tandem is key to building more robust and practical hyperspectral imaging solutions for maize seed variety classification.	Anantha
26	Seed Quality Assurance in Maize Breeding Programs: Tests to Explain Variation	https://access.onlinelibrary.wiley.com/doi/10.2134/agronj2007.0151	This study examined the relationship between standard seed quality tests and actual field emergence performance in maize breeding programs, drawing on data collected across 2002 and 2003 using Standard Germination, Saturated Cold, Soak tests, and the Inbred Quality Index. A central finding of the study was that the tests and index of parental inbreds and F2 progeny correlated poorly with early field emergence of F1 hybrids, raising questions about the practical utility of these measures in hybrid selection. The study further highlighted a methodological concern whereby averaging multiple seed quality tests into a single index risks masking poor individual test performance, potentially leading to misleading quality assessments. While the seed quality index retained value as a ranking tool for comparing inbreds, the findings made clear that it falls short as a reliable selection criterion, and that no single test alone is sufficient, with all tests collectively required for meaningful index calculation.	Anantha
27	Classification of Corn Seed Quality Using Convolutional Neural Network with Region Proposal and Data Augmentation	https://journal.uad.ac.id/index.php/IJTEKI/article/view/26222	Recognising the difficulty of extracting reliable quality information from physical seed images alone, alongside the challenges of lengthy CNN training times and limited dataset availability, this study proposed an enhanced CNN framework incorporating region proposals, Convex Hull processing, and data augmentation for maize seed quality classification. The inclusion of region proposals contributed a 0.3% gain in accuracy, with the model ultimately achieving an average accuracy of 99.01% to 99.33% alongside a remarkably efficient training time of just 2 minutes and 30 seconds. Error metrics further validated the model's precision, with MSE, RMSE, and MAE recorded at 0.0125, 0.118, and 0.0108 respectively, while experimental data testing accuracy ranged between 77% and 99% depending on conditions. The study also acknowledged remaining limitations including the absence of regional and environmental factor considerations, limited pest and disease resistance data, and the need for expanded primary datasets and refinements in feature map design and loss function optimisation to further strengthen model robustness.	Anantha
28	Early Identification of Seed Maize and Common Maize Production Fields Using	https://www.mdpi.com/2072-4292/12/13/2140	Distinguishing seed maize production fields from common maize fields presents a significant challenge due to the near-identical phenological development profiles of the two crop types, a gap this study addressed by leveraging Sentinel-2 satellite imagery combined with machine learning algorithms for early-season field identification. The feature space was constructed using four spectral bands and eight vegetation indices, with KNN, Support Vector Classification, Random Forest, and LSTM evaluated as candidate classifiers. RF and LSTM emerged as the strongest performers, both exceeding 88% accuracy on Sentinel-2 imagery from March to June, with LSTM reaching 90%, while KNN and SVC ranged between 82% and 86%. GF-1 imagery produced comparatively lower accuracy at 85%, and automated sample generation using historical data showed a notable decline in reliability over time, dropping to 74% by September and 83% by July. The study established Sentinel-2 combined with RF and LSTM as a viable and scalable framework for large-scale, early-season monitoring of the seed maize supply chain, while also acknowledging the limitations of automated sampling approaches in sustaining long-term classification accuracy.	Anantha
29	Nondestructive Classification of Maize Moldy Seeds by Hyperspectral Imaging	https://www.mdpi.com/1424-8220/22/16/6064	Mould-affected maize seeds pose a serious threat to germination rates and overall crop quality, yet existing classification methods have largely struggled to assess seed condition without compromising the physical integrity of the sample. This study tackled that limitation by combining hyperspectral imaging with a modified reverse sparrow search algorithm, termed JYSSA, paired with a Random Forest classifier to enable accurate, nondestructive classification of moldy maize seeds. The JYSSA modification was specifically introduced to overcome the tendency of the standard sparrow search algorithm to become trapped in local optima, thereby improving the feature selection process and overall model optimisation. The resulting JYSSA-RF framework achieved a classification accuracy of 96%, a precision of 100%, and a recall rate of 93%, collectively demonstrating that hyperspectral imaging guided by intelligent optimisation algorithms can deliver highly reliable and structure-preserving seed quality classification for practical agricultural applications.	Anantha
30	Purity Identification of Maize Seed Based on Color Characteristics	https://link.springer.com/chapter/10.1007/978-3-642-18354-6_73	Identifying hybrid maize seed purity presents a particularly difficult challenge given the close visual resemblance between legitimate hybrid seeds and female parent self-crosses, further compounded by natural variability in crown fullness, uneven colour distribution, and inconsistencies introduced during artificial placement. This study approached the problem through colour-based image analysis, employing image segmentation alongside RGB and HSV colour models, Fisher discriminant theory, and the K-means clustering algorithm to extract and classify distinguishing colour characteristics from maize seed images. The K-means algorithm operating on one-dimensional feature space derived through Fisher discriminant theory proved effective across four maize varieties including nongda108, ludan981, zhengdan958, and jundan20, consistently achieving a recognition rate exceeding 93.75%. While the study acknowledged imperfections in the clustering features used during experimentation, the results nonetheless demonstrated that colour characteristic analysis offers a practical and accessible method for maize seed purity identification without requiring complex instrumentation.	Anantha
31	A Systematic Review of Machine Learning Based Classification and Prediction Models for Grains (Maize)	https://ieeexplore.ieee.org/document/10961273	A systematic review of 30 papers examining maize seed quality classification paints a clear picture of where the field currently stands and where it still needs to go. The overwhelming majority of studies rely on morphological features related to the size, shape, and structure of seeds which is combined with a variety of imaging techniques to classify seed variety and detect defects. Across this body of work, two broad methodological camps emerge: those using classical machine learning algorithms trained on large datasets, and those leveraging deep learning-based solutions that incorporate feature selection and extraction to let the model identify the most discriminative visual patterns on its own. Both approaches have shown considerable promise, and together they reflect a field that has matured significantly in its ability to automate what was once a manual and error-prone process. However, the review also surfaces a gap that is surprisingly easy to overlook: the automated classification of maize grain quality based on size. Despite size being a practically important quality parameter in seed grading and production, current methods for assessing it remain largely manual, and the research community has yet to develop robust automated solutions that can handle this need at scale. It is a reminder that even in a well-studied area, there are corners of the problem that remain underserved and worth pursuing.	Delina
32	Maturity determination of single maize seed by using near-infrared hyperspectral imaging coupled with comparative analysis of multiple classification models	https://www.sciencedirect.com/science/article/abs/pii/S1350449520396	Determining maize seed maturity accurately and rapidly has remained a persistent challenge in seed quality assessment, with existing practices often falling short in both consistency and technological sophistication. To address this, researchers turned to near-infrared hyperspectral imaging (NIR-HSI), a technique capable of capturing rich spectral information across the seed surface without any physical interference. The study extracted average spectra from both the embryo side (T1) and the endosperm side (T2) of maize seeds, using these as inputs for three classification models which are partial least square discriminant analysis (PLS-DA), decision tree (DT), and adaptive boosting (AdaBoost) with principal component analysis (PCA) applied to identify the most informative feature wavelengths. The results clearly favored the endosperm side: the PLS-DA model built on T2 spectra achieved a perfect classification accuracy of 100%, while the embryo-side model delivered a still-impressive 98.7%, effectively resolving a previously open question about which side of the seed yields more reliable spectral data for maturity classification. These findings make a strong case for NIR-HSI as a rapid, non-destructive, and highly accurate tool for maize seed maturity assessment, and they demonstrate that the choice of spectral acquisition surface matters meaningfully. At the same time, the study points toward the need for further optimization of classification algorithms and broader application of advanced imaging technologies in seed quality workflows, areas where current practice has yet to fully catch up with what the technology is capable of delivering.	Delina
33	Maize seed variety identification model using image processing and deep learning	https://ijeccs.laescore.com/index.php/IJECES/article/view/29715	Identifying maize seed varieties is a deceptively difficult problem where many varieties share strikingly similar morphological features and visual characteristics, making reliable differentiation a challenge that has historically demanded specialized instruments, expert knowledge, and considerable time and expense. Existing methods have often proven too cumbersome for practical deployment, and the application of deep learning to this specific problem has remained comparatively underexplored relative to other grains. To tackle this, researchers developed a hybrid feature-based model that intelligently combines the strengths of multiple techniques: convolutional neural networks (CNNs) for deep feature extraction, histogram of oriented gradients (HOG) for handcrafted feature representation, and Gabor filters for texture analysis, with a multi-class support vector machine (MCSVM) serving as the final classifier. The performance comparison across models told a clear story, end-to-end CNN alone achieved 86%, HOG with SVM reached 90%, CNN with Gabor filter and SVM reached 90.62%, but the full hybrid model combining CNN and HOG features with SVM emerged as the clear winner at 99% accuracy. This result demonstrates that thoughtfully integrating handcrafted and deep features can meaningfully outperform any single approach in isolation, offering a more robust and accessible pathway for maize seed variety identification. Looking ahead, the challenge of classifying overlapping seed images remains an open problem, and continued work in this direction will be essential for building systems that can perform reliably under the messier conditions of real-world seed processing environments.	Delina
34	Maize seeds forecasting with hybrid directional and bi-directional long short-term memory models	https://onlinelibrary.wiley.com/doi/10.1002/fsn3.3783	Maize cultivar classification has long posed a challenge for automated systems, particularly when it comes to developing methods that are both accurate and capable of capturing the complex spatial and sequential patterns embedded in seed images. To address this, researchers constructed a substantial dataset of 14,469 maize seed images and systematically evaluated a range of hybrid deep learning architectures that pair established convolutional networks with temporal sequence models. The study tested combinations of AlexNet and ResNet50 which both are implemented through transfer learning, with LSTM and BiLSTM layers, the latter being capable of processing sequential information in both forward and backward directions to capture richer contextual relationships within the data. Among all configurations tested, the ResNet50+BiLSTM hybrid emerged as the top performer, achieving a classification success rate of 98.10% and demonstrating that combining the spatial feature extraction strengths of ResNet50 with the bidirectional sequential learning capabilities of BiLSTM yields a more powerful classifier than either architecture could deliver independently. While the study is primarily focused on solving the practical problem of maize cultivar classification rather than explicitly mapping out research gaps, its very design implies an acknowledgment that existing single-architecture approaches have fallen short and the results make a compelling case that hybrid models represent a more effective and promising direction for seed quality classification going forward.	Delina

35	Evaluation of mechanical damage in seeds of maize (Zea mays L.) by X-ray and digital imaging	https://www.semanticscholar.org/paper/Evaluation-of-mechanical-dam	Understanding how mechanical damage affects maize seed quality has important practical implications for seed production and handling, yet the relationship between physical injury and downstream seedling development is more nuanced than it might first appear. In a focused study of fifty maize seeds from the AG-510 cultivar, researchers used digital X-ray imaging alongside visual light photography and image processing through Sci-Image software to examine this relationship in detail. The findings revealed that X-ray imaging could effectively link the morphology of embryonic and endospermal tissues to both seed and seedling quality outcomes, offering a window into internal structural conditions that visual inspection alone cannot provide. Crucially, the study found that not all physical ruptures carry equal consequences where damage becomes particularly problematic when it restricts nutrient translocation to the embryonic axis, as is this interference with the seed's internal supply chain that gives rise to abnormal seedling development. This distinction between consequential and inconsequential damage is a meaningful contribution, as it moves the conversation beyond simply detecting injury toward understanding which types of injury actually matter for crop outcomes. That said, the study leaves several questions open: the mechanisms linking specific damage types to quality loss are not yet fully understood, and the exclusive focus on the AG-510 cultivar raises reasonable questions about how broadly these findings can be applied across other maize varieties which is an area that would benefit from further targeted investigation.	Delina
36	Quality Categorisation of Corn (Zea mays) Seed using Feature-Based Classifier and Deep Learning on Digital Images	https://ieeexplore.ieee.org/document/9581464	Sorting good seeds from bad has traditionally relied on manual grading, a process that is inherently subjective and prone to inconsistency, while mechanical alternatives like sieve machines offer only a partial solution, since they are limited to grading by size alone and cannot account for the other visual cues that genuinely indicate seed quality. To move beyond these constraints, researchers developed an imaging-based classification system that evaluates corn seeds using a combination of shape, color, size, and roundness features, comparing several feature-based methods against a deep learning approach built on the ResNet-50 architecture. The results showed a clear hierarchy in performance: feature-based methods such as size and color, and size, shape, and color combined, performed well, with F1 scores mostly above 0.9, while the roundness-based method lagged somewhat behind at 0.854, suggesting it would benefit from incorporating additional descriptive parameters to capture quality more comprehensively. Ultimately, it was the deep learning approach using ResNet-50 that delivered the strongest performance, achieving an impressive F1 score of 0.983 and confirming that learned visual representations can outperform handcrafted feature combinations in distinguishing good seeds from bad. These findings make a strong case for adopting deep learning-based imaging systems in seed grading workflows, offering a more objective, consistent, and accurate alternative to both manual inspection and traditional mechanical sorting methods.	Delina
37	Color and texture for corn seed classification by machine vision	https://ieeexplore.ieee.org/document/6146100	Maize seed quality control stands to benefit considerably from automated classification systems, and one study set out to demonstrate this by developing a machine vision method built around color and texture features rather than relying on deep learning architectures alone. The approach extracted color information using histograms in both RGB and HSV color spaces, paired with texture analysis derived from Grey Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) techniques, with a Support Vector Machine (SVM) serving as the final classifier. Tested across a substantial dataset of more than 10,000 sample images, the system achieved a strong 95.6% accuracy in identifying normal seeds, though performance dropped to 80.6% when classifying defective seeds with gaps that highlights just how much more visually subtle and variable defect patterns tend to be compared to the relatively consistent appearance of healthy seeds. While the study does not explicitly outline the broader research gaps it leaves behind, the disparity in accuracy between normal and defective seed classification quietly points toward an area ripe for further refinement, suggesting that future work could focus on improving feature representation or model design specifically to better capture the more elusive characteristics that distinguish defective seeds from their healthy counterparts.	Delina
38	Spectral and Image Integrated Analysis of Hyperspectral Data for Waxy Corn Seed Variety Classification	https://www.mdpi.com/1424-8220/15/7/15578	While hyperspectral imaging has been used in seed quality assessment before, most prior work focused on spectral information alone, leaving the potential value of combining spectral and spatial appearance data at the level of individual maize kernels largely unexplored. Addressing this gap, researchers applied VIS/NIR hyperspectral imaging alongside a comprehensive feature extraction pipeline that captured both morphological characteristics including area, circularity, aspect ratio, roundness, and solidity with texture features such as energy, contrast, correlation, and entropy. The spectral data itself was refined using Savitzky-Golay smoothing and derivation, with the successive projections algorithm (SPA) employed to select the most informative wavelengths, while classification was carried out using Support Vector Machines and Partial Least Squares-Discriminant Analysis, validated through a leave-one-out cross-validation approach. The results affirmed the value of this fused approach: the SVM model achieved 98.2% accuracy when analyzing the germ side of kernels and 96.3% for the endosperm side, with the combination of spectral and morphological data consistently outperforming either type of information used in isolation. These findings make a compelling case that integrating appearance-based and spectral features captures a more complete picture of seed quality than either approach alone could offer. Looking forward, the researchers point to several promising directions, including building more robust models that incorporate seed samples from different growing regions and years, expanding the spectral range of the hyperspectral imaging system, deepening the morphological and textural characterization of maize kernels, and exploring tighter integration between NIR spectral data and a broader set of image-derived features.	Delina
39	Effect of Sowing Date on Maize Seed Yield and Quality: A Review	https://archive.conscientiabeam.com/index.php/69/article/view/152	The relationship between sowing date and maize seed yield and quality is far from universal, with research showing that its effects differ considerably from one country to another, early sowing proves advantageous in some regions, while delayed sowing works better in others, depending on local climate and growing conditions. This pattern has been documented across numerous studies, where researchers have relied on a mix of field experiments, phenological tracking, growing degree day (GDD) analysis, and direct field observations to understand how timing influences crop performance. Taken together, this body of work generally points toward early or mid-early sowing as the more reliable choice for hybrid seed production, since it gives the crop a longer growth window and exposes it to more favorable environmental conditions, ultimately translating into better yields and higher seed quality. That said, the existing research leaves several gaps that limit how confidently these findings can be applied. There's notably little literature examining sowing date effects specifically under Indian growing conditions, and the results that do exist tend to vary inconsistently across different seasons and locations, making it hard to draw firm conclusions. Comparing yield and yield components across different sowing windows is also methodologically tricky, and the GDD system itself appears to overestimate the thermal time actually needed for grain filling. Beyond this, there's still a need for deeper understanding of how maize's phenological development responds to thermal time as planting gets pushed later, an area where current knowledge remains incomplete.	Delina
40	Identification of Seed Maize Fields With High Spatial Resolution and Multiple Spectral Remote Sensing Using Random Forest Classifier	https://www.mdpi.com/2072-4292/12/3/362	Identifying seed maize fields accurately is essential for reliable seed quality monitoring, and research shows that a random forest classifier built on multispectral data combined with texture features can do this remarkably well, achieving overall accuracy rates of 95.90% and 97.79% along with strong Kappa coefficients of 0.92 and 0.95 across two study areas, Huocheng and Hutubi counties respectively, where producer accuracies reached 93.91% and 97.65%. To arrive at these results, researchers drew on time series vegetation indices such as NDVI, EVI, TVI, RVI, NDWI, and DVI to capture how the crop's spectral signature changed over the growing season, alongside texture features extracted through the grey-level co-occurrence matrix and rotation-invariant local binary patterns, with all modeling and analysis carried out using # and the Weka machine learning suite, and notably, the classifier performed best when six vegetation indices were combined with texture features rather than relying on spectral information alone, this "spectrum plus texture" approach consistently delivering higher overall, user, and producer accuracies, with an average overall crop accuracy of 96.85%, a Kappa coefficient of 0.94, and seed maize specifically showing an average user accuracy of 98.03% and producer accuracy of 95.78%; even so, the approach faces real limitations, since building accurate models depends on large volumes of sample data that aren't always available in every region, the classifier still struggles at times with misclassification and omission errors when distinguishing seed maize from similar crops, and an unbalanced sample structure can hinder model training and drag down accuracy, pointing to a clear need for better strategies to balance sample sizes, adapt algorithms accordingly, and develop improved identification approaches suited to areas with limited samples and to assessments made early in the growing season.	Delina
41	Identification of Defective Maize Seeds Using Hyperspectral Imaging Combined with Deep Learning	https://www.mdpi.com/2304-8158/12/1/144	This paper presents a non-destructive approach for detecting defective maize seeds using hyperspectral imaging (HSI) and deep learning. The authors collected spectral data from healthy and worm-eaten maize seeds and applied preprocessing techniques, including detrending and multiple scattering correction, to enhance spectral discrimination. A feature-selection convolutional neural network (CNN-FES) was used to identify the most informative wavelengths, reducing the spectral dimensionality to 24 key wavelengths, while an attention-based CNN (CNN-ATM) was employed for classification. The proposed model achieved high performance with 97.50% accuracy, 98.28% sensitivity, and 96.77% specificity, demonstrating the effectiveness of combining hyperspectral imaging with attention-based deep learning for maize seed defect detection. However, the study was limited to a small dataset containing only healthy and worm-eaten seeds and was evaluated under controlled laboratory conditions. The framework was not validated across different maize varieties, environmental conditions, or additional defect categories. Future research can focus on expanding the dataset, incorporating multiple defect types, improving model generalization, and integrating advanced self-supervised or transformer-based architectures for practical large-scale deployment.	Ashik
42	A Vision Transformer Network SeedViT for Classification of Maize Seeds	https://onlinelibrary.wiley.com/doi/10.1111/fpe.13998	This paper proposes SeedViT, an improved Vision Transformer (ViT) model for maize seed quality classification. Unlike conventional convolutional neural networks, SeedViT leverages a self-attention mechanism and introduces modifications such as the GeLU activation function and soft image splitting to improve performance on small and medium-sized datasets. The model was evaluated on a dataset of 2,500 maize seed images and achieved an accuracy of 97.6%, sensitivity of 94.1%, specificity of 98.9%, and precision of 97%, outperforming several traditional machine learning and deep learning approaches. The authors also compared ViT-based and VGG-based feature extraction combined with SVM classification, where ViT-SVM achieved a stable accuracy of approximately 96.6%. In addition, visual attention maps were generated to improve interpretability and highlight the regions influencing classification decisions. The study demonstrates that Vision Transformer architectures can effectively address maize seed quality classification with limited training data. However, the model was evaluated on a relatively small dataset and lacks validation across diverse environmental conditions and seed varieties. Future work can focus on large-scale validation, integration with self-supervised and contrastive learning approaches, and the incorporation of advanced attention mechanisms for more robust seed quality assessment and defect detection.	Ashik
43	A Lightweight Method for Maize Seed Defects Identification Based on Convolutional Block Attention Module	https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2023.1153226/full	This paper proposes a lightweight deep learning framework for maize seed defect identification by integrating the Convolutional Block Attention Module (CBAM) with a pretrained MobileNetV3-Large network. The objective is to achieve accurate and efficient non-destructive detection of maize seed defects while maintaining low computational complexity suitable for real-time and mobile applications. The authors utilized a dataset of 12,784 maize seed images comprising seven categories, including normal seeds and six defect types: fusarium, sprouted, moldy, broken, heated, and pest-attacked seeds. Transfer learning from ImageNet and CBAM-based channel and spatial attention mechanisms were employed to enhance feature extraction and improve model convergence. Experimental results showed that the proposed MobileNetV3-Large+CBAM model achieved an overall classification accuracy of 97.95%, a true positive rate of 93.14%, a false positive rate of 1.14%, and an average test time of 35.25 ms, outperforming several popular deep learning architectures while maintaining a lightweight structure. Ablation studies further demonstrated that transfer learning and attention mechanisms significantly improved recognition performance. However, the study was limited to seven defect categories and controlled image acquisition conditions, and the model exhibited reduced performance when distinguishing highly similar defect types. Future research can focus on validating the framework on larger and more diverse datasets, handling multiple simultaneous defects, improving robustness under real-world agricultural conditions, and integrating advanced self-supervised, transformer-based, or multimodal attention architectures for intelligent seed quality assessment and defect detection systems.	Ashik

44	Rapid detection of the viability of naturally aged maize seeds using multimodal data fusion and explainable deep learning techniques	https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2023.1153226/full	This paper proposes a multimodal deep learning framework for rapid and non-destructive detection of viability in naturally aged maize seeds using data collected from multiple sensing modalities, including machine vision (MV), Raman spectroscopy (RS), transmission spectroscopy (TS), fluorescence spectroscopy (FS), and SS sensors. The authors developed the MSCNSVN model, which integrates information from different sensor sources through multimodal data fusion while incorporating explainable deep learning techniques. Experimental results showed that multimodal fusion significantly outperformed single-modal approaches, improving prediction accuracy by approximately 10%, with the combination of MV, RS, and FS data achieving the best performance. Among individual modalities, the fluorescence spectroscopy dataset produced the highest accuracy, while the use of dual-variety and endosperm-surface datasets further improved performance by 2–3%. The study demonstrates the effectiveness of combining multiple sensor modalities and explainable deep learning for maize seed viability assessment. However, the results indicate that model performance is influenced by dataset composition, suggesting the need for broader validation across diverse seed varieties and conditions. Future work can focus on expanding dataset diversity, optimizing multimodal fusion strategies, and improving the generalization of explainable deep learning models for large-scale seed quality monitoring applications.	Ashik
45	Self-supervised maize kernel classification and segmentation for embryo identification	https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2023.1108355/full	This paper presents a self-supervised learning framework for maize kernel classification and embryo segmentation using contrastive learning techniques. The authors employ NNCLR and SimCLR to pretrain deep learning models on unlabeled maize kernel images acquired from a commercial high-throughput imaging system, reducing the dependence on large annotated datasets. The pretrained models are subsequently fine-tuned for classification and segmentation tasks, demonstrating superior performance compared to ImageNet-pretrained and randomly initialized models. Experimental results show that SSL-pretrained segmentation models achieved a DICE similarity coefficient of 0.81, outperforming ImageNet-pretrained (0.78) and randomly initialized (0.73) baselines. The study further demonstrates that competitive performance can be achieved with as little as 1% labeled data, highlighting the annotation efficiency and transferability of self-supervised contrastive learning across multiple downstream agricultural applications. However, the work focuses primarily on kernel orientation classification and embryo segmentation and does not evaluate seed quality assessment, defect detection, or real-world field deployment. Future research can explore the integration of attention mechanisms, seed quality evaluation tasks, and broader validation across diverse agricultural datasets to improve robustness and applicability.	Ashik
46	Identification of Maize Seed Varieties Using MobileNetV2 with Improved Attention Mechanism CBAM	Identification of Maize Seed Varieties Using MobileNetV2 with Improved Attention Mechanism CBAM	This paper proposes I_CBAM_MobileNetV2, an improved MobileNetV2 model integrated with a modified Convolutional Block Attention Module (CBAM) for non-destructive maize seed variety identification. To overcome the limitations of traditional seed classification methods based on texture, shape, and color analysis, the authors collected 3,938 images from 11 maize seed varieties and introduced a parallel attention structure (I_CBAM) by replacing the conventional cascade connection between channel and spatial attention modules. The proposed model achieved an accuracy of 98.21%, outperforming MobileNetV2 by 4.88% and surpassing Xception, MobileNetV3, DenseNet121, E-AlexNet, and ResNet50. Grad-CAM visualization demonstrated that the model effectively focused on discriminative seed features, improving both classification accuracy and interpretability, while maintaining a lightweight model size of 25.1 MB suitable for mobile deployment. However, the study was limited to 11 maize varieties under controlled imaging conditions and did not evaluate performance across diverse environmental conditions or large-scale agricultural settings. Future work can focus on larger and more diverse datasets, advanced attention mechanisms, and extending the framework to seed quality assessment, defect detection, and intelligent seed sorting applications.	Ashik
47	Classification of Maize Images Enhanced with Slot Attention Mechanism in Deep Learning Architectures	Classification of Maize Images Enhanced with Slot Attention Mechanism in Deep Learning Architectures	This paper proposes Slot-Maize, a deep learning framework that integrates CNN, Slot Attention, GRU, LSTM, and Grad-CAM for automated classification of haploid and diploid maize seeds. The Slot Attention mechanism enhances feature extraction by focusing on meaningful seed regions, while GRU and LSTM capture feature dependencies to improve classification performance. The model was evaluated on the Rovlie Maize Seed Dataset and the Maize Variety Dataset, achieving classification accuracies of 96.97% and 92.30%, respectively, outperforming several existing deep learning models. Grad-CAM further improved model interpretability by highlighting the image regions influencing classification decisions. However, the study is limited to seed variety and haploid–diploid classification and does not evaluate multiple defect categories, seed quality assessment, or performance under diverse real-world imaging conditions. Future work can focus on validating the framework on larger and more diverse maize seed datasets, extending the model to identify multiple seed defects and quality attributes, improving robustness under varying environmental conditions, and optimizing the architecture for real-time deployment in automated seed inspection systems.	Ashik
48	DualTransAttNet: A Hybrid Model with a Dual Attention Mechanism for Corn Seed Classification	https://www.mdpi.com/2073-4395/15/1/200	This paper proposes DualTransAttNet, a hybrid deep learning framework for rapid and non-destructive corn seed classification by integrating hyperspectral and RGB image information. To overcome the limitations of single-source imaging and traditional feature extraction methods, the authors combine 3D-CNN for multi-dimensional feature extraction, HV-Attention for capturing global horizontal and vertical image features, and a Transformer Attention module for learning long-range dependencies. The framework performs multimodal feature fusion using hyperspectral and spatial image information, enabling both local and global feature representation. The model was evaluated on a hyperspectral dataset containing six corn varieties and achieved an overall accuracy of 90.01%, an F1-score of 88.9%, and a Kappa coefficient of 88.4%, while maintaining a compact model size of 1.758 MB and an inference time of 0.019 ms, making it suitable for intelligent agricultural applications. However, the study focuses primarily on corn seed variety classification using controlled laboratory hyperspectral imaging and does not evaluate seed quality assessment, defect detection, or performance under diverse environmental conditions and larger real-world datasets. Future work can focus on validating the framework across more maize varieties and field conditions, improving multimodal feature fusion with advanced attention or self-supervised learning techniques, extending the model to detect multiple seed defects and quality attributes, and developing lightweight real-time systems for automated seed sorting and quality inspection.	Ashik